

Unsupervised Discovery and Supervised Prediction of Subsea Production Hydraulic System States from Time-Series

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1. Abstract

Subsea production control systems generate large volumes of multivariate time-series data. This data comes from hydraulic system instrumentation, measuring sensors and devices that monitor hydraulic data, spread across the seabed and platform/topside installations. It contains information about operating conditions and system health. Currently, interpretation is typically manual and reactive. This can delay anomaly detection, increase the risk of production downtime, and risk environmental non-compliance.

The primary motivation for the project is to design a value-adding system that supports operations, enhancing efficiency, production performance, and environmental responsibility. As part of the operator's legislative permit responsibilities ^{1, 2}, it is mandatory that a) annual use and discharge do not exceed permitted volume and b) operators monitor and report use and discharge via the Environmental and Emissions Monitoring System.

This study develops a framework for automatic identification and classification of subsea control system hydraulic operating states. The approach combines unsupervised learning for operating regime discovery with supervised models for system operating state classification.

The main goal is to verify whether unsupervised density-based clustering reliably identifies distinct operating regimes in subsea systems, and if supervised models can subsequently classify these regimes accurately from engineered features.

Using data from the Otter Oil field, hierarchical density-based clustering (HDBSCAN) was applied to identify clusters of system operating regimes. Each cluster was interpreted and labelled using data analysis. Supervised models were then trained on the labelled data for prediction.

These outputs will enable the earlier detection of steady-state operation and abnormal behaviour. They also support targeted maintenance planning and provide auditable evidence for compliance with regulations.

All subsea systems operate similarly, but no two are identical in architecture or operating conditions. This project focuses on a specific field. The same approach is transferable but requires asset-specific tuning.

2. Introduction

2.1 Context

Subsea production control systems (SPCS) monitor and oversee hydraulics, valves, and sensors across wells that can be several tens of kilometres from a host facility. See supplementary Section 2 for an overview of an SPCS.

These systems provide continuous multivariate time-series data from instrumentation to a control system unit. Automated classification of operating regimes — such as steady state, depressurised, or valve operations — is rarely in place. Detecting anomalies, such as external leaks, is typically a reactive, manual process. The risks and possible consequences of anomalies are substantial, including production delays, environmental non-compliance, and costly Dive Support Vessel interventions for repairs. Operators are responsible for monitoring and recording fluid usage for regulatory compliance, as fluid use is regulated. This currently involves manual recording.

2.2 Objectives

Three operational goals:

1. Machine Learning - State identification and Anomaly detection.

Develop ML models that automatically predict hydraulic system operating regimes to detect abnormal behaviours. 5 states:

Steady State, FCV Operation, HP External Losses, LP External Losses, System Depressurised.

2. Data Engineering – Feature Construction and System Health Indicators.

Derive value-added features and system health KPIs from the raw data that reflect system performance and health.

3. Data Visualisation Operational Dashboard.

Display results in a dashboard to support operations in a clear one-page system.

The project workflow for achieving these goals is detailed in Figure 1.

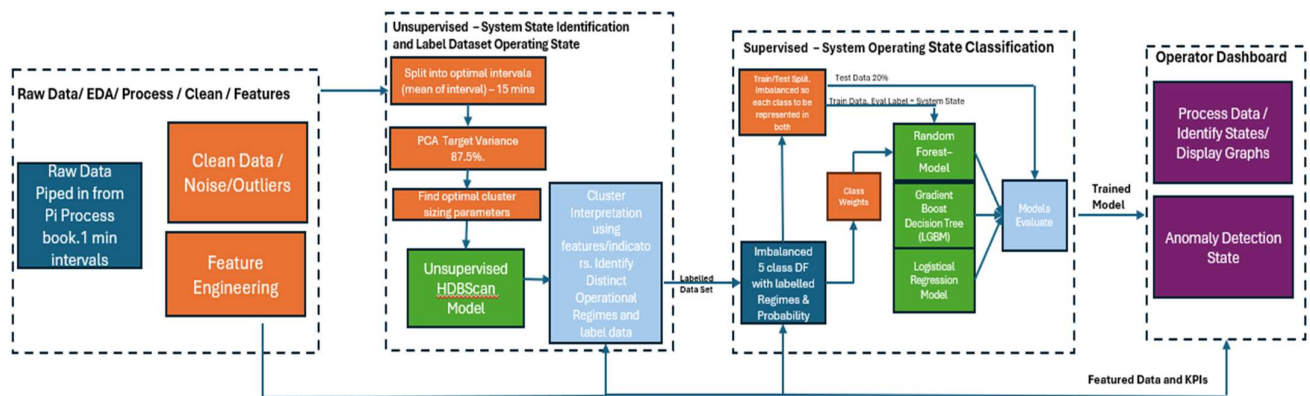


Figure 1. Project Workflow for Hydraulic System State Identification and Classification. The process starts with data EDA, cleaning, and feature engineering, followed by the unsupervised learning phase to identify and label different operating states. The labelled states are then used to train supervised classification models to forecast system conditions. The final stage displays model outputs and system KPIs on an operator dashboard for monitoring and anomaly detection.

3. Data

3.1 Data Source and Nature

The data used in this study were sourced from the Otter Oil field in the UKCS. It was exported from the onshore historian (OSIsoft Process Book) as CSV files at 1-minute resolution ⁴. The dataset contains 4 years of sensor data logged at 60-second cycles from 2020 to 2024.

The raw data covers analogue signals (pressures, flow rates, levels) and boolean valve positions. Raw data variables are detailed in the Supplementary Material, Section 10.

Raw CSVs were converted to Apache Parquet ³ using Python. This conversion compresses files, improving processing efficiency.

3.2 Exploratory Data Analysis (EDA) and Data Processing

Several challenges were identified when dealing with time-series hydraulic control data, requiring intensive data processing and feature engineering, described in Table 1.

Data Challenge	Description	Approach Taken
High dimensionality & correlated variables	70 raw sensor channels, many measuring related parameters, i.e. pressures at different points	Valve types are grouped by type (3.4.1). PCA was applied to reduce dimensionality by transforming correlated features into uncorrelated components while preserving high variance (4.2).
Raw data provides limited insight into system health	The raw data didn't give clear information on the system's health for system monitoring	Feature engineering to derive data points, such as pump runtimes, valve activity, supply usage, external fluid losses, and system status indicators (e.g., whether the system is pressurised). Section 3.3 provides feature details.
Noise, drift, faults & missing values	EDA identified signal noise, instrument drift, communication dropouts, and faulty readings.	Smoothing and rule-based fault filtering.
Temporal dependence	Individual timestamps provide limited insight; patterns occur over time.	Rolling-window ²¹ features and interval-based aggregation to ensure sequence and context are given to learning. $Mu\mu_w(T_j) = \frac{1}{m} \sum_{i=0}^{m-1} s(T_{j-i}) \text{ with } m = \min(w, j+1)$
Class imbalance	Steady state dominates. Abnormal regimes are rare.	Class weighting is used in supervised models to ensure that minority regimes are detected.
Skewed distributions & outliers	Several variables exhibit heavy tails and are prone to extreme events. Supplementary Information Section 8 has details of these variables.	Data features skewness ²⁹ were evaluated using the Fisher-Pearson standardised moment (scipy.stats.skew()), see supplementary section 8. Features with skew > 1 (right-tailed) were log-transformed. $yt = \log(1 + xt)$
Different units & scales	Variables measured in bar, litres, and percentages.	Standardisation ²² of all numeric features before modelling. $x' = \frac{x - \mu}{\sigma}.$
Quantity of data	20 years of 1-minute data made feature extraction and model computation very inefficient.	Used 5 years of data and resampled to 15-minute intervals (mean). A 15-minute timeframe provided the best balance of accuracy, granularity, and computation time.

Unlabelled dataset	The time series contains no predefined labels (ground truths) identifying system operating states. This makes supervised classification impossible without first inferring these states.	Unsupervised clustering (HDBSCAN) was applied to discover distinct operating regimes automatically. These clusters were then interpreted using domain-driven rules and assigned regime labels. The validity of the labels was assessed by comparing regimes with operational data and known system behaviour.
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Table 1 Summary of the data challenges and the approach taken

3.3 Feature Engineering

Supplementary information, Section 6, contains details of the data processing pipelines. Section 5 provides the contents and parameters used in the feature calculations. The process of feature engineering involves creating more suitable and informative features for model training by combining or integrating existing features in new ways²⁷

3.3.1 Valve Operations and Usage

26 valves were bucketed (Supplementary Section 8, variables # 35 to 60) into three types based on their size and fluid consumption on actuation:

1. 2" Low-Pressure (LP) – Annulus Master Valves (AMVs).
2. 5" LP – Production, Water, and Test Valves (PMV, PWV, PDV, TDV, Inlet).
3. High-Pressure (HP) – Surface-Controlled Subsurface Safety Valves (SCSSVs).

Valve transitions were classified as Close-to-Open or Open-to-Close. Only Close-to-Open operations use hydraulic fluid, with usage volumes calculated using valve-specific swept volumes defined in the System Hydraulic Analysis Document⁸.

Four valves (P3_PMV, P3_AMV, P3_PDV, I1_AMV) showed indicator faults, consistently reporting an 'Open' state; these were considered 'Open' while in the Fault condition. All valve operations and timings were recorded in an event log for later dashboard visualisation.

3.3.2 Flow Control Valve (FCV) Usage

The FCV controls hydrocarbon flow and often actuates during unstable production, consuming about 0.4 L per pulse¹³. To prevent false signals caused by $\pm 0.5\%$ noise in the 4–20 mA position sensor, only one full-step transition was recorded. Each confirmed step was multiplied by the known choke step volume from system design data¹³ to calculate fluid usage.

3.3.3 Tank Level Conversion and Smoothing

Tank levels, recorded as percentages, were converted to litres using capacities defined in the HPU schematic¹². The Return Tank Level A transmitter was unreliable; therefore, redundant Return Level B was used throughout.

Tank level signal in per cent p_t .

Tank capacity in litres V_{tank} (constant).

$$L_t = \frac{p_t}{100} V_{\text{tank}}$$

Infrequent noise was observed in both supply and return tank signals, likely due to interference. Several smoothing methods were evaluated, including median filtering, moving averages, and Total Variation (TV) denoising. The TV denoiser^{5,23} was selected for its ability to suppress smaller transient spikes while maintaining the main characteristic behaviour, while removing small, noisy spikes. This ensured the levels accuracy remained when determining fluid consumption.

3.3.4 Umbilical Charge Detection

Substantial hydraulic fluid volumes are consumed during umbilical and accumulator pressurisation following system depressurisation. It is essential to identify these, so they are not misclassified as anomalies. These events were detected from pressure transitions (LPA, LPB, HPA, HPB) and classified as pressurisation if completed within a two-hour window (if longer than this, it may be an anomaly). The charge volume was calculated from the associated drop in the supply tank level and was proportionally distributed over the event duration. A pressurisation event is logged on the dashboard, including the time and volumes.

3.3.5 System Fluid Consumption and Tank Fills

Overall system consumption was derived from the negative gradient of the smoothed Supply Tank Level signal, representing fluid outflow.

Positive gradients are tank fill events, detected as one-minute positive level changes exceeding a defined threshold (to prevent noise from falsely flagging a fill). Fills separated by a defined gap were treated as distinct events, and their cumulative volumes were summed at the end of each event in the dashboard. This approach automates regulatory recording details.

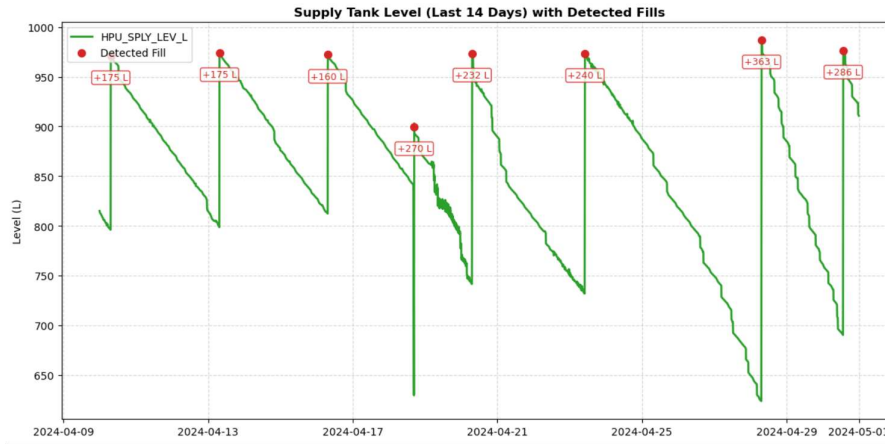


Figure 2: Tank fill events with calculated 'fill' volumes overlaid

3.3.6 External Loss Calculation

External losses, i.e. to sea, were calculated as the difference between total supply consumption (the negative gradient of the supply tank) and calculated usage:

$$\text{External Loss} = \text{Total Consumption} - (\text{Valve Usage} + \text{FCV Usage} + \text{Umbilical Charge Volume})$$

An example occurred in Q1 2020¹³, where the closure of a single valve coincided with a significant reduction in fluid losses.

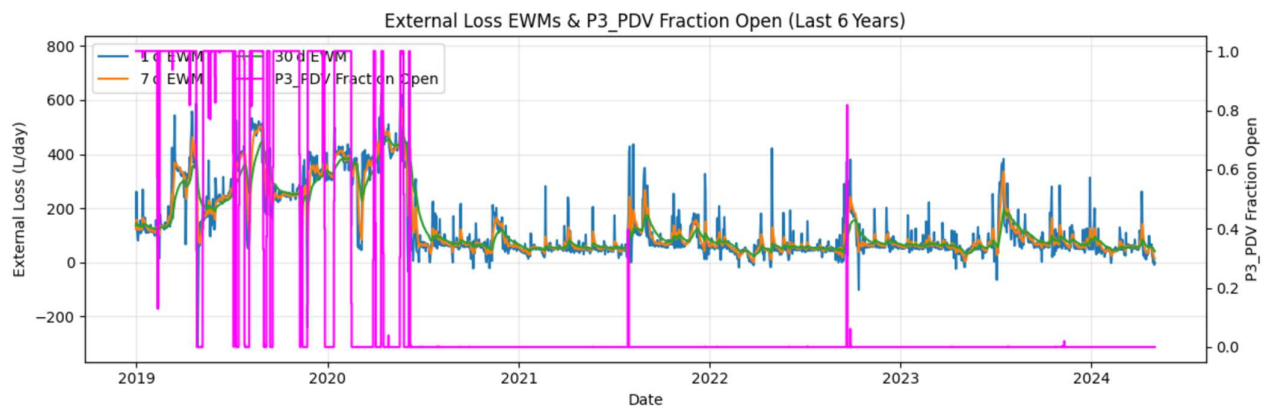


Figure 3 Elevated external losses with the P3 well Production Diverter Valve(magenta) in the Open position(1); the losses are stemmed when the PDV is closed (0).

3.3.7 Baseline Hydraulic Usage

A baseline system usage rate was derived from periods where there was no activity for 12 hours:

This baseline was validated against the Otter Hydraulic Leak Investigation Report¹³, which reported an approximate 40 l/day consistent loss with no operations. These baseline points serve as a reference for monitoring system behaviours.

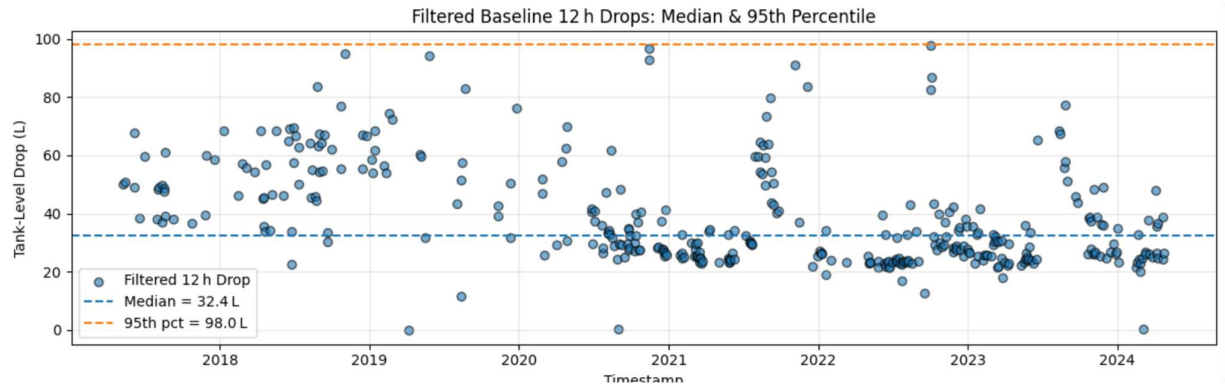


Figure 4 Baseline fluid scatter plot. Each point represents an operationally “inactive” 12-hour period with that period’s daily fluid consumption rate

3.3.8 Pressure Differentials

Pressure deltas between the topside and subsea HP/LP sensors are expected to have an 18-bar offset at a depth of 180 m due to hydrostatic pressure. Any deviations from the delta may indicate a change in system integrity.

3.3.9 Pump Runtime Inference

No pump status signal was present in the dataset, so they were derived. LP/HP runtime periods were derived from the supply pressures using rise detection, enabling pump start/stop identification (at 1-minute resolution). i.e., if pressure increases, the pump runs until the pressure rise ceases. This is vital information for detecting changes in system behaviour and for planning pump maintenance routines based on operational time rather than a set time, e.g., annual maintenance.

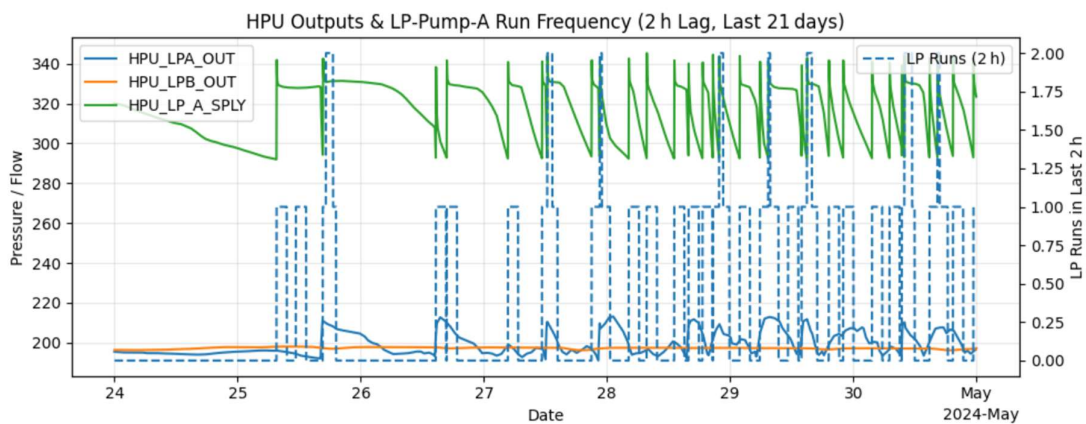


Figure 5: Derived LP Pump Run frequency with the pressure sensor in green and the derived pump run frequency. The pump runs more frequently to maintain the system pressure when the LPA Supply pressure (green) depletes faster.

3.4 Final Dataset Model

The final training dataset used in the model is summarised in Table 2. It consists of thirty-one selected raw variables and engineered features that collectively characterise

the system behaviour. For each variable feature, rolling statistical features (mean and standard deviation) were computed over 30-minute, 1-hour, and 2-hour windows to capture short and long-term temporal dependencies in system behaviour. For variables with skewness >1 , log-transformed values were used. This process yielded 217 features in total before PCA dimensionality reduction.

Feature Group	Raw Column	Transformations Applied
LP System Metrics	HPU_LPA_OUT_DropSlope_1h_Lph,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	HPU_LPB_OUT_DropSlope_1h_Lph,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	HPU_LPA_OUT,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	HPU_LPB_OUT,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	SCM1_RET,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	SCM1_LP_CONS,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	LP_pump_rate_2h,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
HP System Metrics	LP_pump_state,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	HPU_HPA_OUT_DropSlope_1h_Lph,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	HPU_HPB_OUT_DropSlope_1h_Lph,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	HPU_HPA_OUT,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	HPU_HPB_OUT,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	SCM1_HP_CONS,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	HP_pump_state,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
Hydraulic Fluid Consumption and Loss Metrics	HP_pump_rate_2h,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	Supply_Consumption_Excl_Fills,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	Supply_Consumption_Excl_Fills_DropSlope,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	External_Losses,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	External_Loss_MA_2h,	Log transformed Original + Log rolling mean and std (30 min, 1 h, 2 h) = 7 features
	External_Loss_EWM_1d,	Log transformed Original + Log rolling mean and std (30 min, 1 h, 2 h) = 7 features
	External_Loss_EWM_7d,	Log transformed Original + Log rolling mean and std (30 min, 1 h, 2 h) = 7 features
Valve and FCV demand Metrics	External_Loss_MA_12h,	Log transformed Original + Log rolling mean and std (30 min, 1 h, 2 h) = 7 features
	External_Losses_DropSlope_1h_Lph,	Log transformed Original + Log rolling mean and std (30 min, 1 h, 2 h) = 7 features
	FCV_Fluid_Usage,	Log transformed Original + Log rolling mean and std (30 min, 1 h, 2 h) = 7 features
	2_LP_Valve_ClosedToOpen,	Log transformed Original + Log rolling mean and std (30 min, 1 h, 2 h) = 7 features
	2_LP_Valve_OpenToClosed,	Log transformed Original + Log rolling mean and std (30 min, 1 h, 2 h) = 7 features
Umbilical Charging & Pressure Delta Metrics	5_LP_Valve_ClosedToOpen,	Log transformed Original + Log rolling mean and std (30 min, 1 h, 2 h) = 7 features
	5_LP_Valve_OpenToClosed,	Log transformed Original + Log rolling mean and std (30 min, 1 h, 2 h) = 7 features
	umbilical_charge_volume,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	LP_Px_Delta,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features
	HP_Px_Delta,	Original + rolling mean and std (30 min, 1 h, 2 h) = 7 features

Table 2: The selected features from the dataset with transformations used to construct the final training dataset.

Each raw variable is expanded into a 7-dimensional feature vector comprising the original value (log if appropriate) and rolling mean and standard deviation computed over 30-minute, 1-hour, and 2-hour windows.

4. Methods –System Operational Regime Identification and Classification

We deploy a method that combines unsupervised regime discovery, which automatically identifies distinct patterns or groupings (regimes) in the data to label the selected features in the dataset (detailed in Table 2), with supervised regime classification, which uses the now-labelled data to classify these patterns into known operating states. The aim is to transform multivariate time-series data into clear operating regimes, after which a predictive model can automatically classify system states and detect anomalous states.

4.1 Conceptual Overview

The approach consists of two main stages.

1. Unsupervised Operating State/Regime Discovery – identify operating states in data to label the data by regime.
2. Supervised Regime Classification – train predictive models to recognise these states from the data.

4.2 Dimensionality Reduction

Before unsupervised clustering, the dataset's dimensionality was reduced using Principal Component Analysis (PCA)^{7,24,25}. PCA is a linear transformation method that identifies the principal components of the data, capturing the maximum variance in lower-dimensional representations²⁶.

PCA transforms the original data features into a smaller set of uncorrelated components that capture the primary data structure. The number of principal components was set to 48 (see Figure 6 for rationale). This accounted for 87.5% of the data's variance. This reduced the feature space's dimensionality from 217 to 48 while preserving most of the structure, creating a good balance between reduction and information retention.

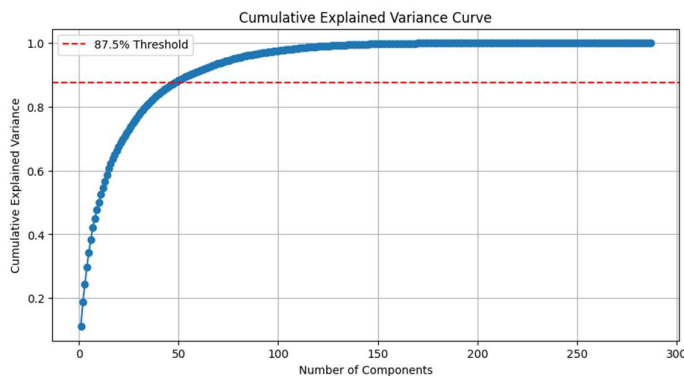


Figure 6. The cumulative explained variance curve for the PCA decomposition rises steeply for the first 48 components before flattening, indicating diminishing returns from additional components and little benefit. The dashed red line marks the 87.5% variance threshold (equal to 48 components). After this, the curve shallows.

4.3 Unsupervised HDBSCAN Regime Identification

An HDBSCAN ^{7 15} model was fitted in the PCA-reduced space to reveal the operating-state clusters.

HDBSCAN estimates the density of each point's neighbourhood and uses this information to build a hierarchy of connected dense regions. Clusters are the parts of this hierarchy that remain stable across different density levels.

HDBSCAN was chosen over DBSCAN because it handles variable density and noise more effectively. DBSCAN requires a single global density threshold, whilst HDBSCAN constructs a density model that allows identification of clusters with varying densities. This is essential for imbalanced data sets. It also provides the probability of cluster confidence. This can be used in a) reviewing probabilities that a cluster belongs to a regime and b) supervised learning to weight the models and ensure they learn well from rarer regimes with high confidence.

In this study, clusters represent consistent patterns of system behaviour, while points that do not fit any cluster are regarded as noise.

A range of cluster configurations was tested to find the best balance that reduces noise, preserves good regime separation, and identifies rarer clusters, while maintaining a level of generality.

The HDBSCAN model requires five key parameters.

- k (number principal components) – The PCA stage reduced dimensionality to the minimum number of components required to explain 87.5% of the data variance
- min cluster size, minimum number of points required for a region to be considered a cluster.
- min_samples determines how strictly density is defined. It specifies the minimum number of neighbours required for a point to be considered a core point.
- The metric defines the distance measure used to compute neighbourhood density. Euclidean distance used as PCA produces an uncorrelated feature space.
- cluster_selection_epsilon is a parameter that relaxes cluster boundaries and merges clusters if they fall within a specified distance threshold. This parameter was set to 0 to maintain clear distinctions between operating regimes and prevent merging rare/small but potentially significant states.

Details of the optimally tuned HDBSCAN parameters are provided in Supplementary Section 7.

4.4 Mapping Clusters to Physical Regimes

Once the clusters were identified, their numerical parameters were analysed to determine their physical meaning and assign each cluster to an operating regime. This interpretation was performed using rule-based criteria that compared cluster statistics with rules summarised in Table 3.

Regime	Regime Criteria
Steady State	Supply consumption (excluding fills) and external losses < 0.04 L/min (58 L/day).
Low-Pressure System Losses	External losses > 0.4 L/min, LP pump rate high, HP pump rate low.
High-Pressure System Losses	External losses > 0.3 L/min, LP pump rate high, HP pump rate low.
System Depressurised	LP and HP pressure sensors < 1 Bar. i.e. at 0 and off
Flow Control Valve Operation	FCV fluid usage > 0.2 L/min and is_fcv > 0.

Table 3: Operating Regime Identification Rules used to identify the operating regime of a cluster

These criteria establish a robust framework for labelling clusters to a regime.

Thresholds for key variables were selected using a combination of historical engineering reports on the system¹³ and data-driven analysis. Sections 4.4.1, 4.4.2 and 4.4.3 explain how the rule thresholds were determined, using histograms and empirical cumulative distribution functions (ECDFs) to identify the boundaries between normal and abnormal behaviour.

Thresholds were validated by analysing their separation across the clusters to verify that each threshold produced clear separation, confirming that the regime boundaries correspond to operational differences. Supplemental information section 3 provides boxplots for all features used to evaluate the clusters.

4.4.1 External Losses

The threshold was chosen near the point where the data frequency drops sharply, marking the start of the tail of the non-steady state. This aligns with the 85% percentile of the baseline fluid.

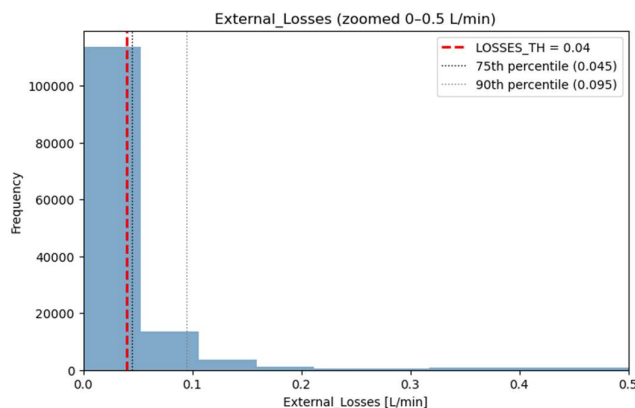


Figure 7 Histogram of External Losses with a threshold of 57 litres per day

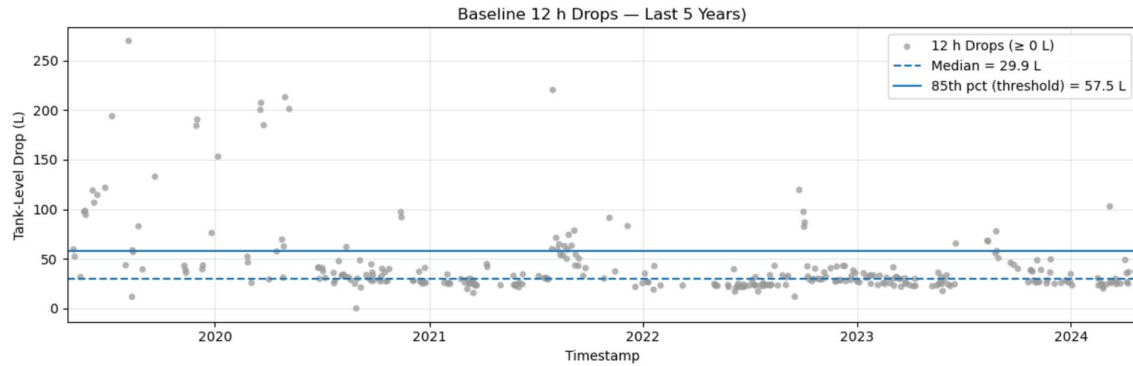


Figure 8: An individual 12-hour tank-level drop occurs when there is no system demand. Each point represents a daily total fluid consumption. The 85% percentile is 57l per day.

4.4.2 FCV Fluid Usage.

This corresponds to approximately the 97th percentile, separating typical idle conditions from frequent FCV operation events. 0.2L is roughly the volume of an FCV pulse¹³

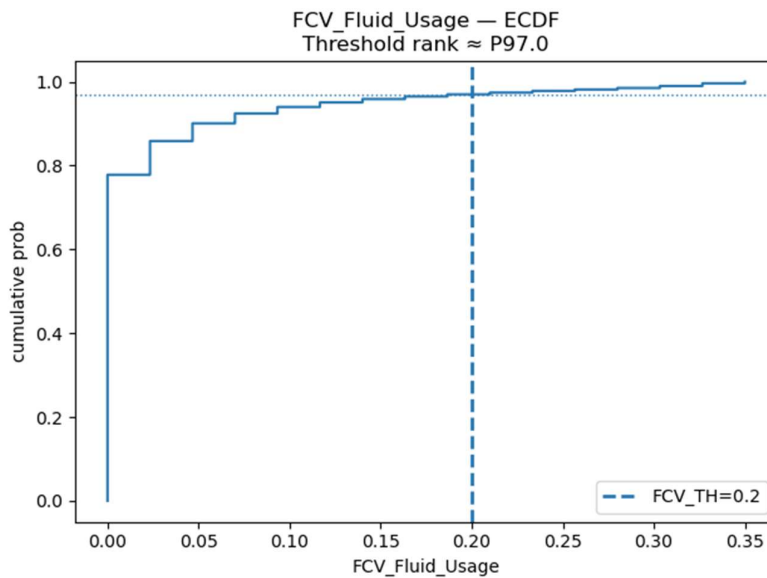


Figure 9 ECDF for FCV_Fluid_Usage. The 0.20 L/min threshold (97th percentile) distinguishes regular idle periods from very active FCV operation events.

4.4.3 HP and LP pump rates:

The threshold was set at the 90th percentile, which represents periods of high usage. Anything above this may be anomalous.

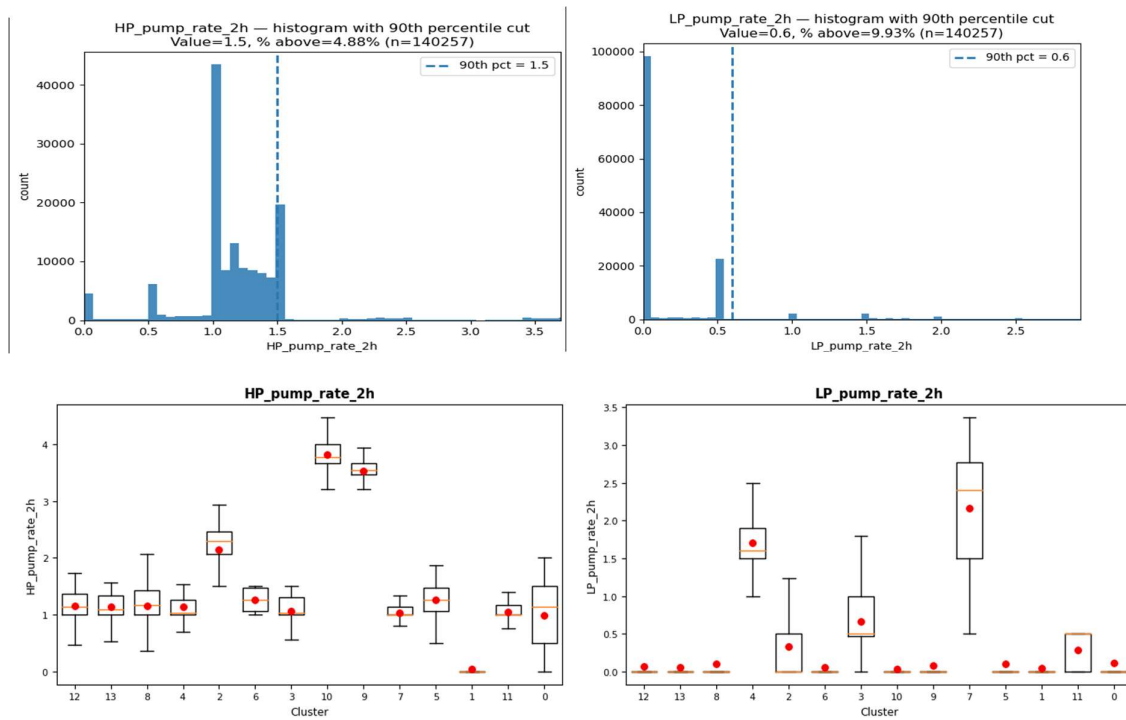


Figure 10 HP and LP Pump Rate Distributions and Boxplots for Pump Threshold Selection. Histograms (top) show the frequency distributions of HP and LP pump rates, with 90th percentile thresholds used to define elevated pump activity. Boxplots (bottom) illustrate the variation in these rates across the clusters.

4.5 Supervised Regime Classification

The cluster-derived regime labels, along with their probabilities (HDBSCAN confidence that they belong to a cluster), are then used to train supervised models that automatically recognise these states in newly provided data. Three models are evaluated:

- LightGBM (LGBM) ⁶ — a gradient-boosted decision-tree ensemble optimised for large datasets. It models complex, non-linear relationships and feature interactions.
- Random Forest (RF) ⁷ – A RF is a series of parallel decision trees. i.e. a bagging ensemble of trees providing a robust baseline performance and interpretability. May generalise better but underfit more complex patterns.
- Logistic Regression (LR) ⁷ estimates class probabilities with a logistic function for multi-class problems. It assumes linear separability and may miss complex interactions, but it is stable and transparent for feature influence.

Because the dataset is highly imbalanced and some cluster labels are uncertain, a single weight is assigned to each training sample. This weight combines two effects:

- a class-balance weight that gives more importance to rare regimes and less to the steady-state regime (represents 95% of the data points)
- a cluster-confidence weight based on HDBSCAN's membership probability, so uncertain samples influence the model less.

The combined weight is :

$$w = (\text{cluster confidence})_i \times (\text{class-balance weight})$$

The weights are only applied to LR and LGBM models. RFs do not naturally incorporate sample weighting.

Cluster confidence is derived from HDBSCAN's cluster probability. The model should learn more strongly from rarer classes with high confidence. The class balance weight gives higher importance to rare states and lower weight to common states.

The training and test data had to be split to ensure regimes were in both datasets. This was achieved by dividing each class into two groups (80% and 20%) and using the last 20% of each group for the test set.

4.6 Data Visualisation and Operator Dashboard

An interactive dashboard was developed in Streamlit²⁰ to provide an operator interface for data visualisation and analysis. The application executes the three data-processing pipelines and the LR classification model, displaying analogue and indicator data (system health status parameters provided in supplementary section 9).

Operators can dynamically select parameters, adjust timeframes, and modify scales to explore system behaviour in detail.

The Logistic Regression model is integrated into a prototype dashboard for operator use.

The system can classify each 15-minute operating state, highlight anomalies, and display KPIs derived from the feature engineering process.

This dashboard will improve the detection of abnormal system behaviour, improve targeted maintenance planning, and improve environmental compliance reporting.

Supplementary information Section 11 provides more detailed images of the dashboard's functionality.

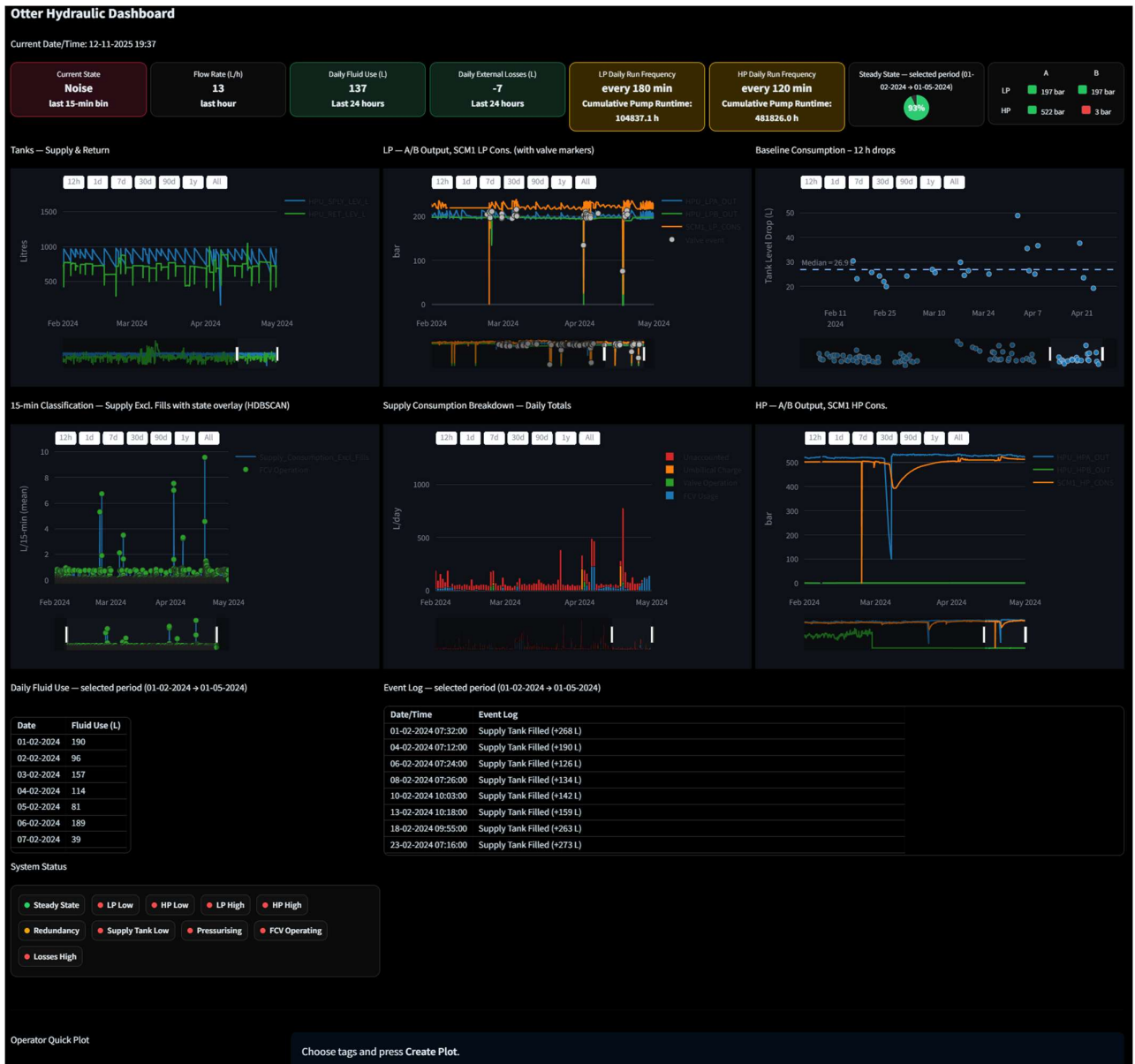


Figure 11 Streamlit Dashboard. Single Page with operating regime model detection, system health indicator KPI, relevant time configurable data plots, event logs with time of tank fills, valve ops, system events/regime change.

5. Results

5.1 Unsupervised Regime Identification

HDBSCAN applied to the PCA-reduced feature space identified twelve stable clusters. 83% of samples were confidently assigned to a single cluster, while the remaining 17% were classified as noise-like or transitional.

```
[PCA] k=48 | X_pca shape=(140257, 48) | cumulative variance=0.875
n_clusters: 12
noise_pct: 17.676
mean_cluster_prob (non-noise): 0.920074
top_cluster_sizes: {11: 83548, 9: 18820, 2: 2554, 5: 2356, 8: 1424, 4: 1289, 3: 1215, 6: 1214, 10: 1169, 7: 710}
```

Figure 12. HDBSCAN results summary. 12 clusters were found. 17.7% points labelled noise i.e. ambiguous/transitional and not confidently assigned to a cluster. For non-noise points, the mean membership probability was high as 92%. Cluster 11 is by far the largest cluster

Detailed HDBSCAN cluster membership confidence and cluster silhouette scores are available in Supplementary Information Section 12.

From the 12 clusters, 5 meaningful physical operating regimes were identified using the set of defined rules (Table 3) to connect the clusters to interpretable operational states.

Two-dimensional PCA projections provided visual validation of the states, confirming that the identified regimes correspond to physically distinct operating states.

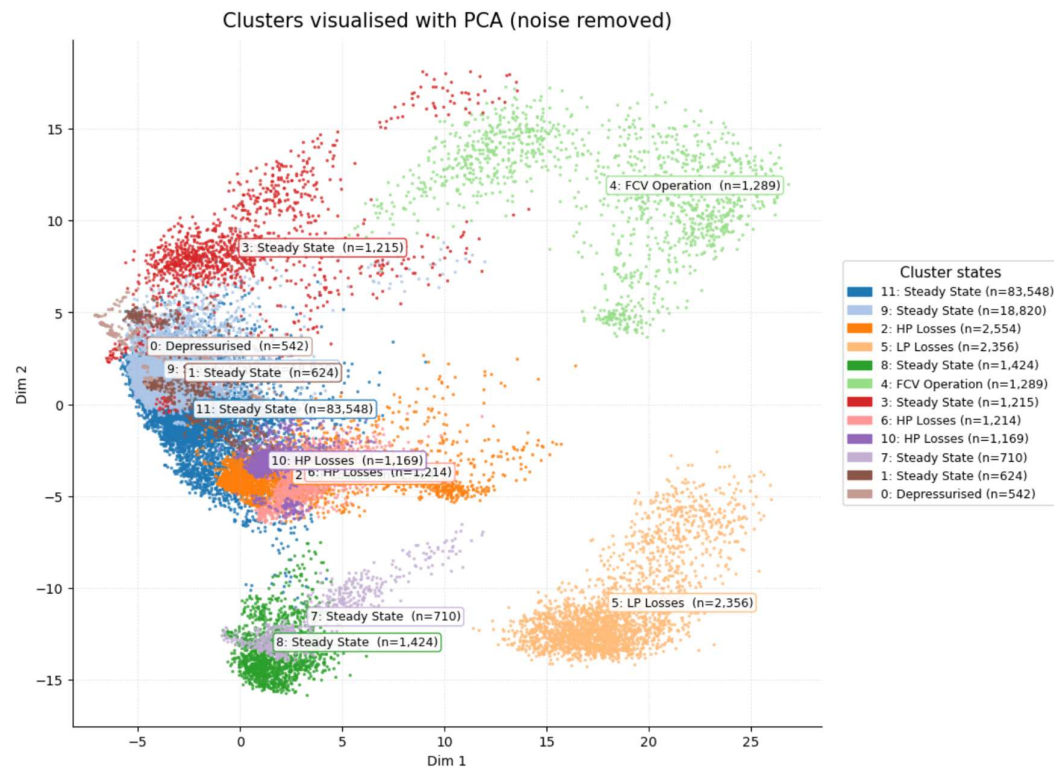


Figure 13 PCA projection of the dataset onto the first two principal components, visualising the discovered operating-state clusters (after noise removal). Each coloured region represents a cluster with its assigned state and sample size.

The HP losses are projected to be close to steady-state points in PCA space, even though their local density structure is distinct enough for HDBSCAN to differentiate them as a separate cluster. This is because the overall system's behaviour is dominated by the LP hydraulic circuit, where the most significant variability occurs. HP events involve smaller volumes and create more subtle deviations from steady state.

The clusters were plotted against the operational data in Figure 14, showing alignment between cluster assignments and the system's fluid consumption. This indicates that the identified clusters correspond to distinct hydraulic operating states at appropriate times. This validates that the applied method effectively distinguishes meaningful hydraulic system states.

Periods of HP losses coincide with higher HP pump rates, comparatively low LP pump rates, and increased overall fluid consumption, aligning with expected behaviour during HP system losses. These three HP loss events (brown) over the dataset time period are visible in Figure 14.

An LP loss regime (orange) aligned with the period pre Q2 2020, when the LP pump showed consistent high demand while the HP pump was idle, again with increased fluid consumption. This period was known to suffer LP fluid losses (see Figure 3).

The majority of identified clusters represent steady-state behaviour, with low consumption rates and low pump demand. During steady-state periods, short transient spikes occur that likely correspond to normal operations (e.g., valve movements, start-ups, pressurisation, instrument noise). The applied framework demonstrates good generalisation, correctly avoiding labelling these spikes as anomalies, reducing false positives.

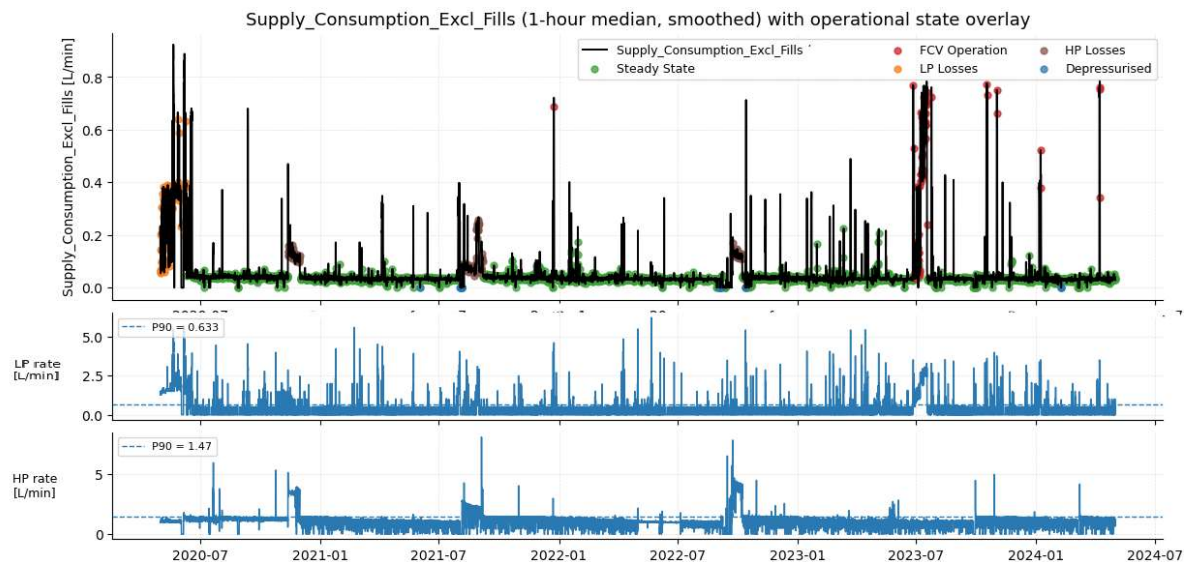


Figure 14 shows the black line. raw fluid consumption, while the coloured markers indicate the dominant operational regime for each one-hour interval. The lower panels display the LP and HP pump-rate distributions, providing additional detail for identifying LP or HP consumption anomalies associated with these operating states.

A scatter plot of external losses versus pump rates also shows that the clustered, labelled operational regimes are effective. Supplementary Information Section 3 contains all whisker box plots to evaluate each cluster's feature points.

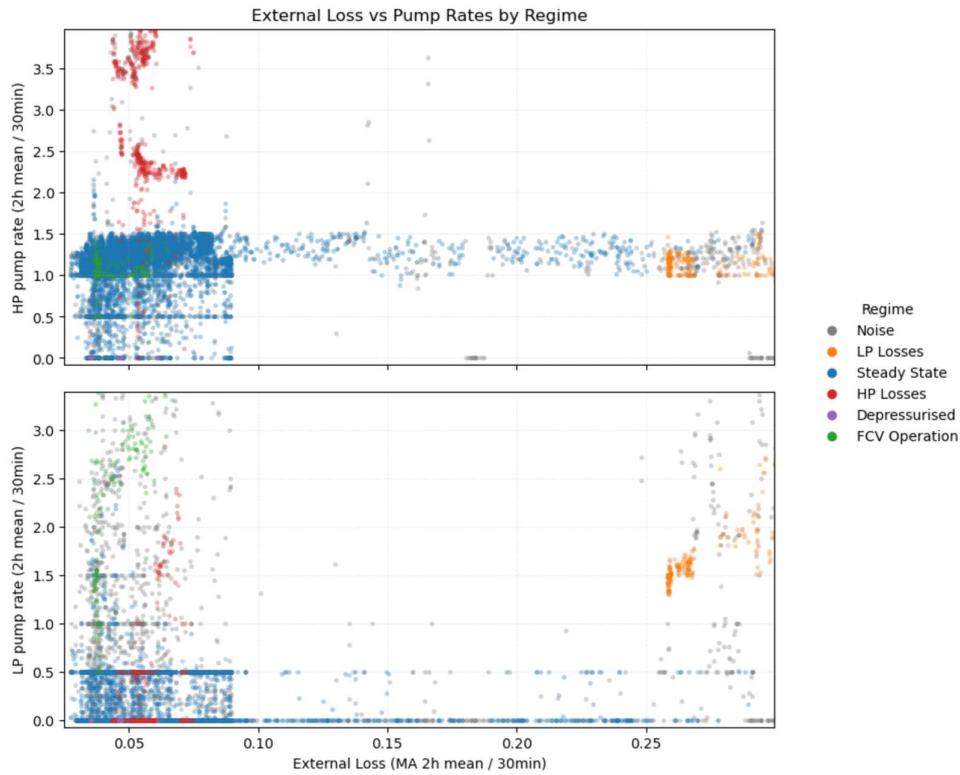


Figure 15: Relationship between external fluid losses and LP/HP pump rates, with points coloured by operational regime, highlighting distinct behavioural patterns. The bottom plot shows a cluster of LP Losses regimes (yellow) with high pump rates and high external losses. FCV operation temporarily increases LP pump rates but does not cause significant losses, as expected. The top plot reveals two distinct clusters of HP losses with high HP pump rates. HP losses have higher HP pump activity; they are minor and less frequent than LP losses. Low losses and steady pump characteristics are evident during the steady periods.

5.2 Supervised Model Interpretation and Validation

The supervised models accurately identified the operating regimes and performed similarly. LR achieved nearly perfect precision and recall, with some errors in the rarer LP Losses regime (mistaken for normal steady-state or FCV operation).

A simple Logistic Regression model can classify these regimes with very high accuracy.

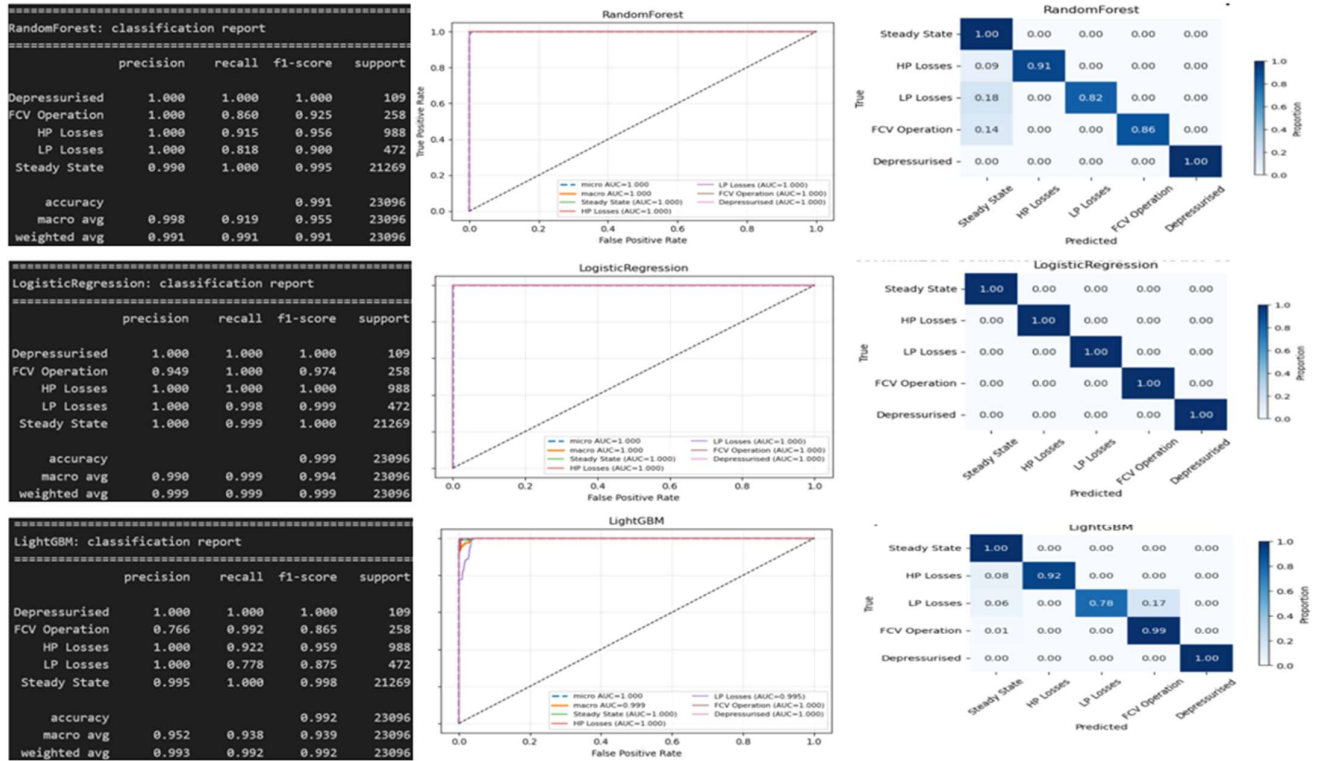


Figure 16 Supervised Model Comparison for Regime Classification. Classification reports, ROC curves, and normalised confusion matrices for Random Forest, Logistic Regression, and LightGBM models

6. Conclusions and Discussion

Similar studies ^{17,19,26,30} in this field have mainly used the same controlled and labelled hydraulic system timeseries dataset, for which supervised techniques are suitable for classifying timeseries data from a hydraulic circuit, and it is possible to measure the accuracy of supervised or unsupervised models against ground truth labels. The main challenge for this project is that, in real life, time-series hydraulic datasets aren't labelled, and operating regimes may not be clear-cut, with transition periods, and anomalies can be rare and short-lived. Dong ²⁷ acknowledges this challenge and notes that obtaining accurate unsupervised models for hydraulic systems “*necessitates a solid understanding of the system*”.

For 83% of that dataset, the project successfully identified and classified distinct operating regimes, including anomalous states. Using unsupervised learning, unlabelled time-series datasets were clustered into groups that closely matched known operational states evident in the data.

The remaining 17% of the data was not confidently assigned to a cluster and was not classified.

The model identified multiple steady-state subgroups. An exploratory investigation determined that these corresponded to different operational configurations (e.g., changes in redundancy but steady state nonetheless). These findings show that even subtle shifts in system configuration can be detected through unsupervised data-driven analysis and modelling.

Notably, the model not only identified high losses but also distinguished between those from the LP and HP circuits (these operate independently).

As the dataset lacks ground-truth labels for operating states, the accuracy of unsupervised regime identification cannot be quantified numerically. Validation relied on cluster membership probabilities, temporal alignment with known hydraulic behaviour and domain knowledge. These assessments proved that the inferred regimes are entirely plausible, but do not provide absolute accuracy values.

Supervised models trained on the labelled, non-noise dataset achieved almost perfect accuracy in predicting operational regimes. Weighted classes based on cluster membership confidence and class rarity improved recognition of the rarer non-steady-state regimes. The near-perfect performance prompted checks for potential data leakage; however, none was found. These results can be explained by the fact that once the unsupervised stage identifies well-separated clusters, a supervised model trained on the high-confidence, non-noise samples can reliably learn and reproduce those same distinctions. The high accuracy on test data may yet fail on unseen anomalies and new operating conditions.

A Graph Neural Network (GNN)²⁸ was trialled during the supervised classification phase of the project. GNNs are suited for data that can be represented as graphs, where entities are nodes and their physical relationships are edges. This structure fits well a hydraulic control system comprising interconnected sensors, valves, and components with physical linkages. The GNN was discounted due to its higher computational cost, yielding no better results than those of other supervised models. See section 4 of the supplementary information for the GNN configuration and results.

This project demonstrates that a combined unsupervised and supervised learning approach can identify and predict the operating regimes of a hydraulic control system from multivariate time-series data.

The data processing pipelines, clustering methodologies, and supervised classification methodologies are transferable to other assets with fine-tuning to their own distinct architectures, faults, and issues.

6.1 Limitations and Future Work

Systems are dominated by steady-state data, which limits the number of examples available for rare anomaly states. This dataset system had examples of the main types of subsea system anomalies and learnt well. Not all systems will have anomalies to label. Future work could explore the use of synthetic data generation to address the lack of anomaly data. Hydraulic simulation software can accurately replicate system behaviour and be used to introduce synthetic failures into a dataset.

It is also true that new operating states/anomalies may arise that the model has not encountered. A complementary approach would be to incorporate One-Class Classification (OCC) methods suggested by Machado and Murano¹⁹. OCC models are trained only on examples of a single behaviour, usually the “normal” class, and then identify deviations from this baseline as anomalies. Integrating OCC after the clustering stage could provide additional benefits. The unsupervised regime discovery model yields high-confidence regions of steady-state behaviour; these could serve as precise training data for a one-class model. The OCC could be used to identify unseen anomalies not present in the training data, i.e., anomalies the classifier has not learned. The OCC component could improve the overall system’s robustness by detecting behaviours that the supervised model does not identify. This may also resolve the challenge of 17% of noise data points that could not be confidently assigned to a cluster, ensuring that any non-steady state is identified as an anomaly of unknown class. The authors further improved OCC performance by introducing regime-specific OCC models trained separately on each cluster.

A key limitation of the thesis method is that it assigns each time period to one dominant state. However, a single period may have multiple operating regimes²⁸, e.g., both LP and HP losses. To overcome this, a multi-label classification approach is needed to assign

multiple labels to a given time point. Molinero Ranera¹⁷ demonstrates that hydraulic system cycles can be linked to various simultaneous states, and that multi-label models can effectively learn these relationships. Orosnjak & Ramos¹⁸ employ multi-feature analysis to identify overlapping degradation across different system aspects using multilabel machine learning. These findings suggest that multiple operating regimes can be labelled within the same time period.

Rather than clustering the dataset and then labelling the clusters after manual data analysis and interpretation, a semi-supervised learning approach could be adopted. A small subset of the data is labelled with known, confident operational states, and clustering is then conducted guided by these limited labelled examples, allowing the model to learn from both labelled and unlabelled data. This could improve the interpretability and accuracy of cluster identification (including the noise/transition points), reducing reliance on post-labelling analysis. Dong²⁷ demonstrated that semi-supervised methods consistently outperform unsupervised methods for anomaly detection in hydraulic condition monitoring when labelled data is scarce. Dong had the advantage of a labelled controlled dataset as input to the semi-supervised model.

A further extension of this for my study, in line with the Dong²⁷ findings, would be to use HDBSCAN to identify operating states within the unlabelled dataset and to assign labels only to points with high cluster membership probabilities (see supplementary material section 12 for cluster membership confidence results). This would produce a smaller but reliable set of labelled examples derived from operational data. These higher-confidence labels could then be the input to a semi-supervised learning model, allowing it to learn operating states from both labelled and unlabelled samples. Such a framework would be very appropriate for hydraulic system monitoring, where labelled datasets are not readily available in practice.

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Unsupervised Discovery and Supervised Prediction of Subsea Production Hydraulic System States from Time-Series

Ross Thomson, 52318940, 7th December 2025

SUPPLEMENTARY MATERIAL

1. Tools Used

The project was conducted in Python, utilising its community packages.

I selected Visual Studio Code as the IDE because of my familiarity with the tool. The environment was set up using the Anaconda package to access and manipulate Python libraries. GitHub was used to manage versions and notebooks, and to control pipelines.

Pandas and NumPy were employed for data reading, cleaning, and manipulation.

Pandas provided the PyArrow engine for handling and processing Parquet files.

Matplotlib and Plotly were used for graphs and data visualisation. Matplotlib generated static plots suitable for analysis and reporting, while Plotly provided interactive exploratory analysis and dashboard integration in conjunction with the Streamlit library.

Scikit-learn ⁷ supplied a comprehensive suite of machine learning algorithms and preprocessing utilities. It was used for scaling (via StandardScaler), dimensionality reduction (PCA), skewness measurement, silhouette score measurement, and supervised learning models such as Random Forest and Logistic Regression.

LightGBM was imported for the LGBM supervised model.

HDBSCAN library was utilised for unsupervised clustering and operating regime discovery. Its hierarchical, density-based approach is efficient for identifying clusters of varying shapes and sizes.

Scipy provided statistical and signal-processing utilities, including correlation analysis. I also trialled using frequency-domain analysis (FFT) to explore temporal and periodic patterns in the time-series data.

Streamlit (and Plotly) were used to develop an interactive dashboard for presenting results and visualising operating regimes. Streamlit provides a framework for deploying a visual dashboard as a web application, and Plotly delivers dynamic, interactive plots.

Pathlib, os, and sys were employed for file handling and processing.

2. Subsea Control System Description and Operation

A subsea control system enables operators on a central platform to monitor and control subsea wellheads located many miles away. It provides real-time data on pressure, temperature, and flow rates, and allows the platform (Topside) to operate remote valves/chokes safely and efficiently produce hydrocarbons. A block diagram of the Otter system schematic^{9,10,11,12,14} is given in figure 1.

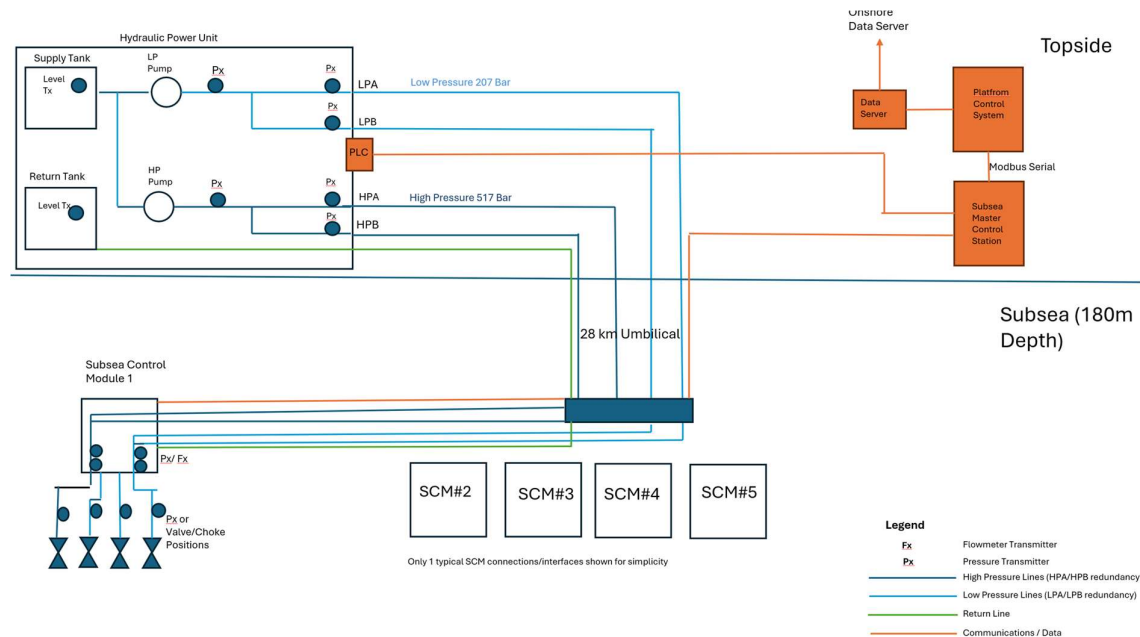


Figure 1 The Otter Field Subsea System Hydraulic System Block Diagram shows the line split, demonstrating equipment located on the topside (platform or FPSO) or subsea (on the seabed at 180m water depth).

The System Type is Multiplexed Electro-Hydraulic. This architecture allows multiple SCMs (typically one per well) to share common communication, electrical, and hydraulic lines through a single umbilical, reducing complexity and cost while enabling control.

- **Master Control Station (MCS)**
The MCS is a topside computer system that sends commands and receives data from subsea equipment. It communicates with the microprocessor in each Subsea Control Module (SCM) via electrical and fibre optic signals.
- **Subsea Control Module (SCM)**
Each SCM is mounted on a wellhead and contains:
 - An electronic module for communication with the MCS.
 - Hydraulic components to operate valves and chokes.
 - Pressure and flow transmitters that provide feedback to the MCS
The SCM also manages multiplexing logic to route hydraulic fluid to the correct valves.

- Hydraulic Power Unit (HPU)

The HPU supplies pressure and fluid to the SCMs/wells.

- The HPU supplies hydraulic fluid in 2 independent hydraulic circuits, a Low Pressure (LP): 207 bar and a High Pressure (HP): 517 bar.
- There are 2 reservoirs. A Supply reservoir that holds clean hydraulic fluid ready for use and a return reservoir: Collects fluid returned from subsea.
- Pump Circuits:
The LP and HP pumps draw fluid from the supply reservoir and discharge into a common header equipped with pressure transmitters.
- Pressure transmitters provide feedback for pump control (start/stop) to maintain system pressure at 207 or 517 bar.
- Redundant Supplies:
From the common header, fluid splits into LPA/LPB and HPA/HPB circuits for redundancy. If one line fails subsea, the other maintains system operation and production.

- Umbilical and Communication

The umbilical contains hydraulic hoses for LP and HP fluid, chemical supplies and the Electrical cables and fibre optics for power and communication.

The umbilical connects the platform topside equipment to SCMs on the seabed.

- SCM Functions

Reads pressure and flow data at regular intervals. Routes hydraulic fluid to valves and choke actuators using solenoids and multiplexing software

3. Cluster Box Plots - Feature Evaluation

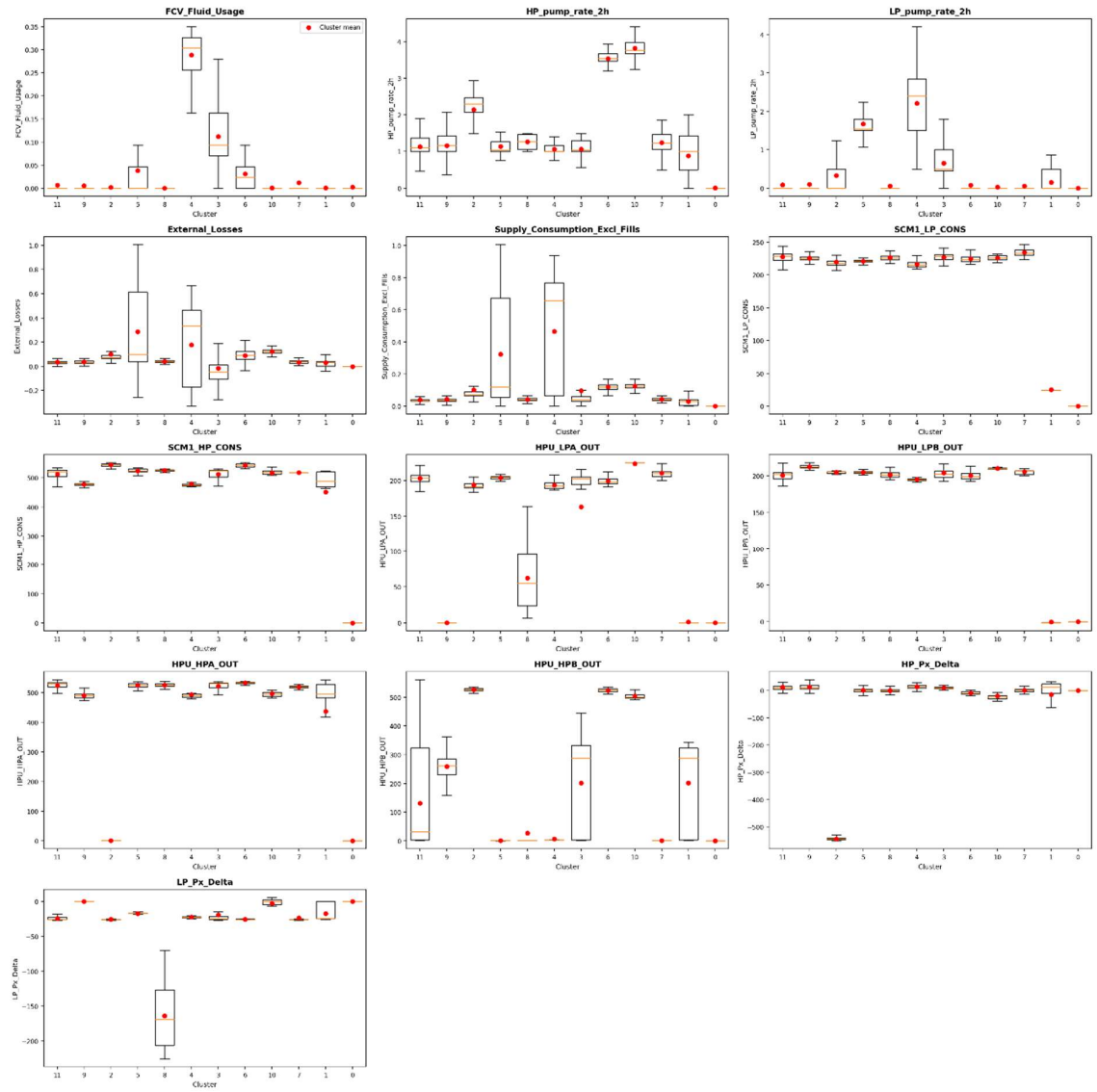


Figure 2 Box plots of feature points for each cluster are used to evaluate which operational regime the HDBSCAN cluster belongs to.

4. Graphical Neural Network

A GNN was implemented using the PyTorch library, with the graph topology defined by to the hydraulic circuit Figure 2.

The model achieved strong performance, comparable to the RF, LightGBM, and the LR classifiers. However, training was computationally intensive, with some runs taking up to 12 hours to complete. Given the computational time and cost, and the fact that simpler supervised models provided nearly perfect results, the GNN was not included in the final evaluation for this thesis. In summary, the results suggest that a GNN based approach could be valuable in future work or projects where hydraulic system structured relationships are present.

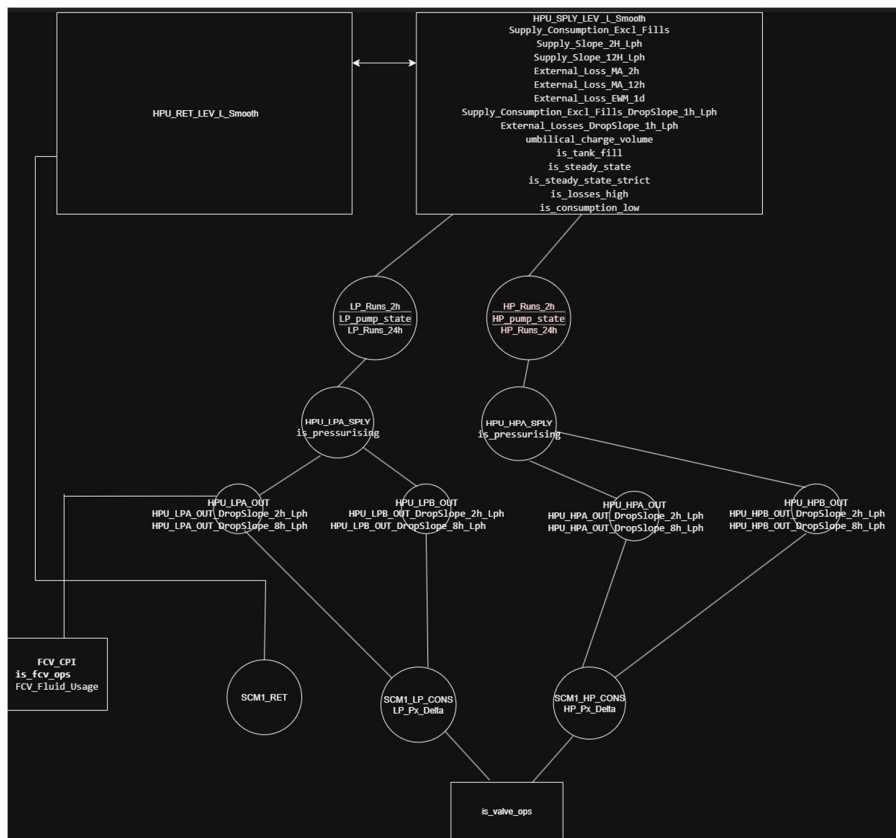


Figure 3. GNN Architecture for the Otter System with nodes and edges which are in line with the physical hydraulic circuit

5. Constants used in Feature Engineering Pipelines

These constants and thresholds were used in the feature engineering pipeline and were saved as `otter_config.py`.

3.1 Physical tank and valve parameters

The following fixed volumes and system properties were used in pipeline calculation, values taken from the reference documents ^{8,12,13,14}:

- Tank capacities
 - Supply tank: 1000 L (`supply_tank_volume`)
 - Return tank: 1800 L (`return_tank_volume`)
- Valve actuation volumes (approximate hydraulic volumes per operation)
 - 5" production/wing/annulus valves: 8 L per actuation (`valve5_volume`)
 - 2" valves: 2 L per actuation (`valve2_volume`)
 - HP valve actuation volume (including line volume): 1 L (`valve_hp_volume`)
- Control valve step volumes
 - Flow control valve (FCV): 0.4 L per step (`fvc_step_volume`)
 - Pressure control valve (PCV): 0.05 L per step (`pcv_step_volume`)
- Hydrostatic pressure
 - Approximate hydrostatic pressure at 182 m water depth: 18.27 bar (`hydrostatic_p`)

3.2 Thresholds for umbilical charging and pressurisation checks

Dictionaries define lower and upper thresholds for detecting umbilical charging and pressurisation events, where the value must be below the lower threshold, rise above it, and reach the higher limit to indicate an event has occurred.

- Umbilical/consumption thresholds (UMBILICAL_THRESHOLDS)
 - SCM1_LP_CONS: 40–190
 - SCM1_HP_CONS: 40–450
 - HPU_LPB_OUT: 30–180
 - HPU_HPB_OUT: 30–440
 - HPU_LPA_OUT: 20–180
 - HPU_HPA_OUT: 20–440

3.3 Valve state encoding

Valve states are mapped from a string to numeric code.

"OPEN" = 1

"CLOSED" = 0

"FAULT" OR "No Data" OR "TIMEOUT" OR "UNKNOWN" = -1

6. Data Feature Processing Pipelines

```
def run_full_pipeline(
    parquet_path: str,
    columns_to_remove: list,
    valve_columns: list,
    valve5_volume: float,
    valve2_volume: float,
    valve_hp_volume: float,
    pcv_columns: list,
    thresholds: dict,
    pump_events_path: str,
    valve_transition_cols: list,
    drop_slope_data_path: str,
    use_smoothed_tank: bool = True,
    smoothing_strength: float = 4.8,
    combine_sensors: str = "mean",
    years_back: int = 8
) -> pd.DataFrame:
    df = load_and_clean_other_data(parquet_path, columns_to_remove, years_back=years_back)
    df = process_valve_data(df, valve_columns)
    df = valve_fluid_usage_calc(df, fluid_per_5lp=valve5_volume, fluid_per_2lp=valve2_volume, fluid_per_hp=valve_hp_volume)
    df = process_fcv(df)
    df = process_pcv(df, pcv_columns)
    df = convert_tank_levels(df)
    df = add_smoothed_supply_level_litres(df, a_l="HPU_SPLY_LEV_L", b_l="HPU_SPLY_LEV_B_L", combine=combine_sensors, strength=3.9, out_col="HPU_SPLY_LEV_L_Smooth")
    df = add_smoothed_return_level_litres(df, a_l="HPU_RET_LEV_L", b_l="HPU_RET_LEV_B_L", combine=combine_sensors, strength=3.9, out_col="HPU_RET_LEV_L_Smooth")
    df = process_umbilical_charges(df, channel_thresholds=thresholds)
    df = system_fluid_consumption(df, supply_col="HPU_SPLY_LEV_L")
    df = add_external_losses(df)
    df = add_external_loss_moving_averages(df)
    df = add_daily_ewm_to_minutely_df(df, loss_col="External_Losses", spans=[1, 7, 30])
    tank_col_to_use = "HPU_SPLY_LEV_L_Smooth" if use_smoothed_tank else "HPU_SPLY_LEV_L"
    df = add_baseline_columns(df, lp_col="SCH1_LP_CONS", hp_col="SCH1_HP_CONS", tank_col=tank_col_to_use, fcv_steps_col="FCV_FullSteps", valve_transition_cols=valve_transition_cols,
    df = add_pressure_deltas(df)
    df = add_pump_run_cumulatives(df, pump_events_path, pumps=("LP_Pump_A", "HP_Pump_A"))
    df = add_slope_features(df, drop_slope_data_path) # Just got LP and HP
    df = detect_fills_safe(df, level_col="HPU_SPLY_LEV_L", min_event_litres=10.0, step_noise=0.5, max_gap="3min", max_step_litres=150.0, smooth_window="5min")
    df = add_pump_runtime_metrics(df, run_col="Cum LP Pump A Run Count", prefix="LP")
    df = add_pump_runtime_metrics(df, run_col="Cum HP Pump A Run Count", prefix="HP")
    df = generate_valve_event_log(df, valve_event_map)
    return df
```

Figure 4 Numeric Feature Data pipeline 1 implemented in python used to clean, process data and create/derive numeric features

```
def run_labelling_pipeline(
    df: pd.DataFrame,
    channel_thresholds: dict,
    pressurising_level_col: str = "HPU_SPLY_LEV_L",
    plot_bool = False,
) -> tuple[pd.DataFrame, dict]:
    metas: dict = {}

    # 1) Low fluid consumption
    df = add_low_fluid_consumption_label(df, col="Supply_Consumption_Ext_Fills", mode="episode", enter_lpm=63/24/60, exit_lpm=65/24/60, min_enter_min=2, min_exit_min=2, out_col="is_consumption_low", return_mask_only=False)

    # 2) Valve + FCV operations
    df, metas["valve_pcv_ops"] = label_valve_pcv_operations(df, valve_col="Valve_Operation_Fluid", pcv_col="PCV_Fluid_Usage", threshold=0.1, threshold_unit="l_per_min", min_on_minutes=1, bridge_gap_min=1, out_col="is_valve_ops", plot=plot)

    # 3) FCV operation mask + flag
    metas["fcv_ops"] = df = label_fcv_operation_from_col(df, col="FCV_Fluid_Usage", threshold=0.3, threshold_unit="l_per_min", min_on_minutes=1, bridge_gap_min=1, out_col="is_fcv_ops")
    df, metas["fcv_ops_simple"] = add_fcv_operation_simple(df, col="FCV_Fluid_Usage", threshold=0.3, out_col="FCV_Operation")

    # 4) Tank fills
    df, metas["tank_fills"] = label_tank_fills(df, level_col="HPU_SPLY_LEV_L", out_col="is_tank_fill", main_period=120, min_threshold=30.0, backup_period=90, backup_threshold=25.0, min_on_minutes=3, bridge_gap_min=1, return_summary=True)

    # 5) Pressurising / umbilical charging
    df, charge_events, press_meta = detect_and_label_pressurising(df, channel_thresholds=channel_thresholds, level_col=pressurising_level_col, window=pd.Timedelta("2H"), min_fluid=0, out_col="is_pressurising", add_volumes=True, vol_col="umbilical_charge_volume")
    metas["pressurising"] = press_meta
    metas["pressurising_events"] = charge_events

    # 6) High external losses
    df, metas["losses_high"] = add_high_external_losses_label(df, col="External_Losses", mode="episode", enter_lpm=32.5/1440, exit_lpm=31.5/1440, min_enter_min=2, min_exit_min=1, out_col="is_losses_high")

    # 7) High / low pressure flags
    df = label_high_pressure(df, lp_col="SCH1_LP_CONS", lp_threshold=4245, hp_col="SCH1_HP_CONS", hp_threshold=4377, out_lp_col="is_lp_high", out_hp_col="is_hp_high", min_true=2, bridge=1)
    df = label_pressure_low(df, lp_col="SCH1_LP_CONS", hp_col="SCH1_HP_CONS", lp_threshold=170, hp_threshold=4480, out_lp="is_lp_low", out_hp="is_hp_low")

    # 8) Redundancy / low supply tank
    mask_or, df = detect_loss_redundancy(df, threshold=100.0, plot=False)
    df["is_no_redundancy"] = mask_or
    mask_low_tank, df = detect_low_supply_tank_level(df, threshold=250.0, plot=False)
    df["is_supply_tank_low"] = mask_low_tank

    # 9) Pump states
    df, metas["pump_states"] = add_pump_states(df, logic_elev="on", logic_high="on", persist_elev_min=30, persist_high_min=30, exit_ratio=0.90, exclude_fill_col="is_tank_fill", floors_hp=(1.2, 1.6), floors_lp=(0.25, 0.75))

    return df, metas
```

Figure 5 Boolean health indicators Feature pipeline 2 used to derive key system statuses from the numeric data. The KPIs are displayed in the operator dashboard.

7. HDBSCAN Best Parameter Fit Results

Experiments were conducted with varying cluster sizes to determine the optimal fit running the HDBSCAN on the data set for each parameter configuration and evaluating optimal results.

The three parameters to fine-tune were :

- k (number principal components) – Targeted to give the minimum number of components/dimensionalities to provide a data variance of 87.5%. Forty-eight of the first PCs were selected.
- min cluster size, identifying rare cases is essential, so kept all clusters with at least 500 points (0.3% of overall data). i.e. for anything to be a regime, it must have at least 500 points.
- min_samples was set to 50, meaning at least 50 neighbouring points are required for a location in feature space to be considered sufficiently dense to form a cluster.

These parameters achieved a balance between reducing noise, preserving good regime separation, and identifying rarer clusters, but maintaining a level of generality. This would provide 12 explainable clusters with minimised noise, accounting for 17.6% of data points (indistinguishable data points).

k	cum_var	min_cluster_size	min_samples	n_clusters	noise_pct	mean_prob
48	0.875	500	50	12	17.676	0.920
48	0.875	500	30	13	27.226	0.881
48	0.875	750	30	11	28.002	0.870
48	0.875	1000	30	10	27.485	0.840
48	0.875	1000	50	13	45.910	0.826
48	0.875	750	50	13	45.910	0.806

Figure 6 HDBSCAN runs to identify optimal HDBSCAN parameter settings.

8. Skewness Testing of Features

Skewness tests²⁹ the asymmetry of a distribution. The study utilised the Fisher–Pearson standardised moment coefficient to evaluate features skewness distribution:

$$\gamma_1 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2\right)^{3/2}}$$

where x_i are the observations, $\bar{x}$ is the mean, and n is the sample size.

This statistic measures how much the distribution deviates from symmetry:

$\gamma_1 = 0$: symmetric distribution

$\gamma_1 > 0$: right skewed

$\gamma_1 < 0$: left skewed

Several raw hydraulic features exhibited strong right sided skew.

Figures 7 and 8 show this behaviour for two dataset features. The log transform reduced skewness where the skew score was greater than 1, spreading values more evenly. Reducing distributional skew is essential to prevent extreme values from dominating the variance, distorting clusters.

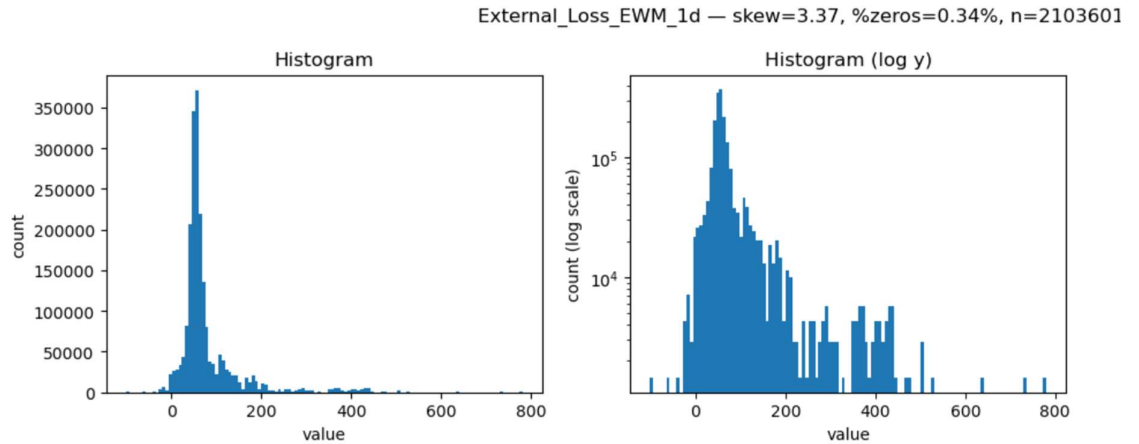


Figure 7: The left side histogram shows the raw distribution, which is strongly right-skewed (skew = 3.37) with a long upper tail. The right panel shows the same data under a log transform, compressing the tails and producing a more symmetric, compact distribution.

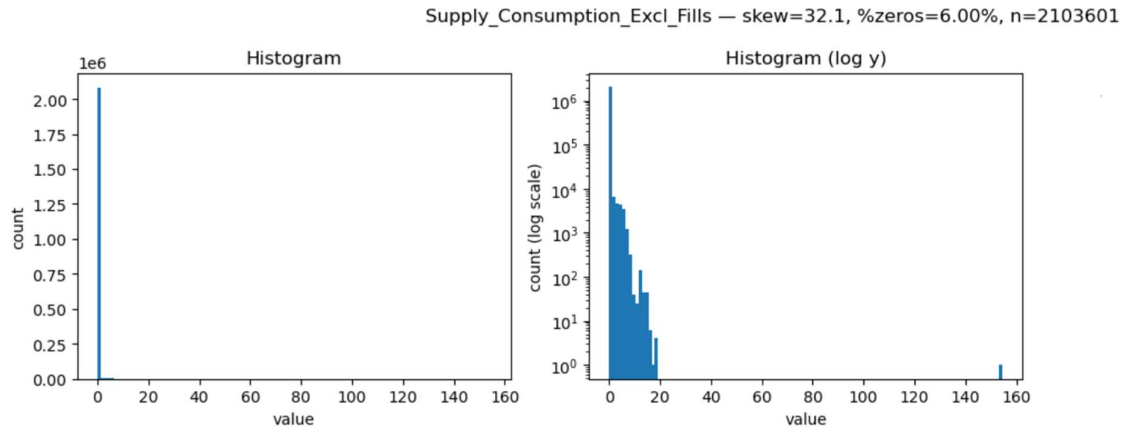


Figure 8 The left panel's raw distribution shows extreme right skew (skew = 32.1), indicating a heavy-tailed distribution with rare but significant events. The right panel shows the same data under a log transform, spreading the data, reducing the influence of larger values and promoting the underlying data structure.

9. System Health Indicators

A series of useful system KPIs was created to improve monitoring of system performance and health. These were not utilised in the model due to potential data leakage, and the desire was to use only numeric data. However, these parameters were displayed on the operator's dashboard because they are extremely useful to the operator.

- `is_valve_ops`: valve demand active if either `Valve_Operation_Fluid` or `FCV_Fluid_Usage` exceeds a threshold.
- `is_fcv_ops`: flow control activity active if `FCV` activity > 1 step
- `is_tank_fill`: detects if supply tank being filled.
- `is_pressurising`: pressure-delta crosses up from low to high within a window, indicating the system is being pressurised.
- `is_losses_high`: When $\text{External_Loss_MA_2h} \geq \theta_{\text{enter}}$ For $\geq N$ minutes. It exists only when the value is below a threshold.
- `is_lp_high` / `is_hp_high`, `is_lp_low` / `is_hp_low`: fixed alarm thresholds (e.g., $\text{LP} \geq 245 \text{ bar}$ / $\leq 145 \text{ bar}$; $\text{HP} \geq 537 \text{ bar}$ / $\leq 350 \text{ bar}$).
- `is_no_redundancy`: if any HPU outlet < 100 bar.
- `is_supply_tank_low`: if tank level < 250 L.
- `HP_pump_state` & `LP_pump_state` (3-level state): cumulative counter that increments by one each time the pump starts, counts the number of runs over 2/24h periods, and classifies the pump runtime as: 1 = Steady, 2 = Elevated, 3 = High.

10. Raw Data Listing

	Variable Name	Unit	Interval(s)	Cleaning/QC/Processing
1	HPU_SPLY_LEV	%	60	Intermittent noise. Smoothing
2	HPU-RET_LEV	%	60	Transmitter Failure, replaced with redundant HPU-RET_LEV_B,
3	HPU_SPLY_LEV_B	%	60	Intermittent noise. Smoothing
4	HPU-RET_LEV_B	%	60	Intermittent noise. Smoothing
5	HPU_LP_A_SPLY	bar	60	
6	HPU_LP_B_SPLY	bar	60	
7	HPU_HP_A_SPLY	bar	60	
8	HPU_HP_B_SPLY	bar	60	
9	HPU_LPA_OUT	bar	60	
10	HPU_LPB_OUT	bar	60	
11	HPU_HPA_OUT	bar	60	
12	HPU_HP_B_OUT	bar	60	
13	HPU_SSIV1_OUT	bar	60	
14	HPU_SSIV2_OUT	bar	60	
15	SCM1_LPA	bar	60	
16	SCM1_LPB	bar	60	
17	SCM1_LP_CONS	bar	60	
18	SCM1_HPA	bar	60	
19	SCM1_HP_B	bar	60	
20	SCM1_HP_CONS	bar	60	
21	SCM1_RET	bar	60	
22	SCM1_LP_FLOW	l/min	60	
23	SCM1_HP_FLOW	l/min	60	
24	SCM1_LP_RET_FLOW	l/min	60	
25	SCM2_LPA	bar	60	
26	SCM2_LPB	bar	60	
27	SCM2_LP_CONS	bar	60	
28	SCM2_HPA	bar	60	
29	SCM2_HP_B	bar	60	
30	SCM2_HP_CONS	bar	60	
31	SCM2_RET	bar	60	
32	SCM2_LP_FLOW	l/min	60	
33	SCM2_HP_FLOW	l/min	60	
34	SCM2_LP_RET_FLOW	l/min	60	
35	P3_PMV	0/1/-1	60	
36	P3_PWV	0/1/-1	60	Position Failure in Open position, so -1 status overridden Open
37	P3_AMV	0/1/-1	60	Position Failure in Open position, so -1 status overridden Open
38	P3_SCSSV	0/1/-1	60	
39	P3_PDV	0/1/-1	60	Position Failure in Open position so -1 status overridden Open
40	P3_TDV	0/1/-1	60	
41	P1_PMV	0/1/-1	60	
42	P1_PWV	0/1/-1	60	
43	P1_AMV	0/1/-1	60	

44	P1_SCSSV	0/1/-1	60	
45	P1_PDV	0/1/-1	60	
46	P1_TDV	0/1/-1	60	
47	P2_PMV	0/1/-1	60	
48	P2_PWV	0/1/-1	60	
49	P2_AMV	0/1/-1	60	
50	P2_SCSSV	0/1/-1	60	
51	P2_PDV	0/1/-1	60	
52	P2_TDV	0/1/-1	60	
53	I1_PMV	0/1/-1	60	
54	I1_PWV	0/1/-1	60	
55	I1_AMV	0/1/-1	60	
56	I2_PMV	0/1/-1	60	
57	I2_PWV	0/1/-1	60	
58	I2_AMV	0/1/-1	60	Position Failure in Open position so -1 status overridden Open
59	MPMV_Inlet	0/1/-1	60	
60	Man_CI	0/1/-1	60	
61	SCM1_LP1_COV	1 or 2	60	Encoded to "LP1" = 1 and "LP2" = 2
62	SCM1_HP_COV	1 or 2	60	Encoded to "HP1" = 1 and "HP2" = 2
63	FCV_CALC	Integer	60	
64	FCV_CPI	%	60	
65	P1_PCV	%	60	
66	P2_PCV	%	60	
67	P3_PCV	%	60	
68	I1_PCV	%	60	
69	I2_PCV	%	60	

11. Operator Visualisation Dashboard

The interactive dashboard created using streamlit and plotly, is designed to provide clear, concise and relevant system status details to support the operator



Figure 9 Derived System Status KPIs. The supervised model identifies the current system operating regime and calculates flow rates, daily fluid usage, and pump run times.

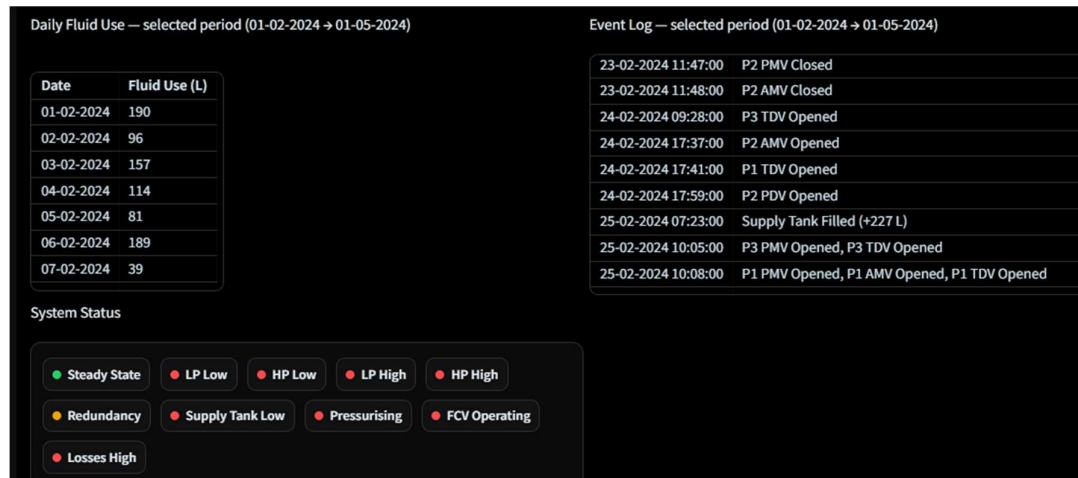


Figure 10. a) A daily fluid usage log for the selected period b) an event log detailing tank fills, valve operations, system pressurisation, etc. c) KPIs derived from the numerical raw data (section 9).

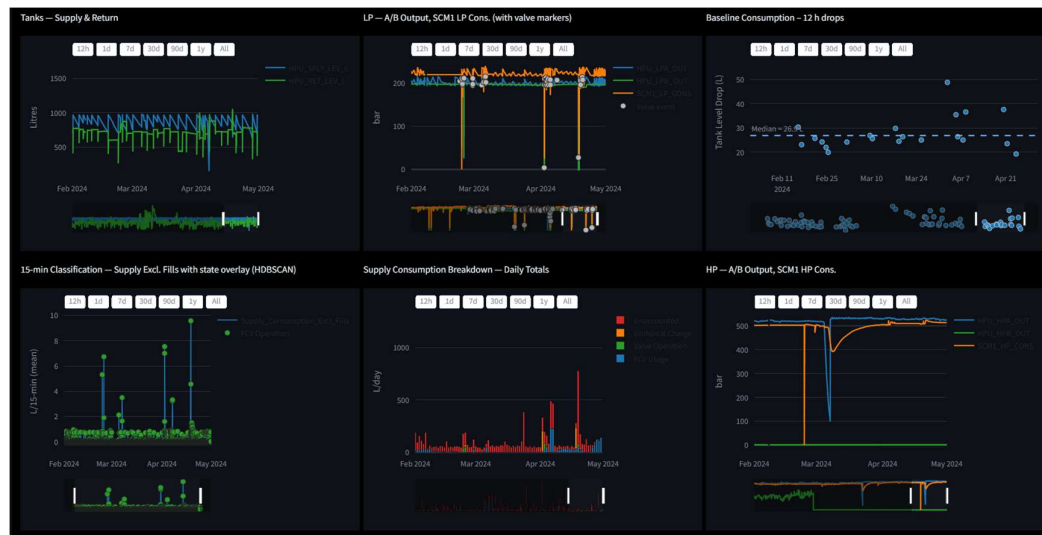


Figure 11. Relevant interactive system plots of feature data for monitoring system status

12. Unsupervised HDBSCAN Membership Confidence Summary and Cluster Quality

The Scikit-learn silhouette score provides a measure of cluster quality, i.e., how well separated and well-formed the clusters are. It combines cluster cohesion (how close each point is to other points in the same cluster) and cluster separation (how far each point is from the nearest point in a different cluster).

The silhouette coefficient³¹ for a point i :

$$s_i = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

Where: $a(i)$ = average distance to its own cluster, $b(i)$ = nearest cluster distance

The score can be interpreted as follows:

- +1: Perfect clustering — the point is far from other clusters and well matched to its own
- 0: Borderline clustering — the point is equidistant between two clusters.
- 1: Poor clustering — the point is closer to another cluster than its own.

The project HDBSCAN silhouette score was 0.22, indicating moderate separation between clusters in the PCA space. This can be expected for hydraulic system time series where regimes overlap, and transitions are commonplace.

HDBSCAN membership probability informs us how confident the method is that a point belongs to its assigned cluster. The project HDBSCAN membership results are summarised in Figures 12 and 13.

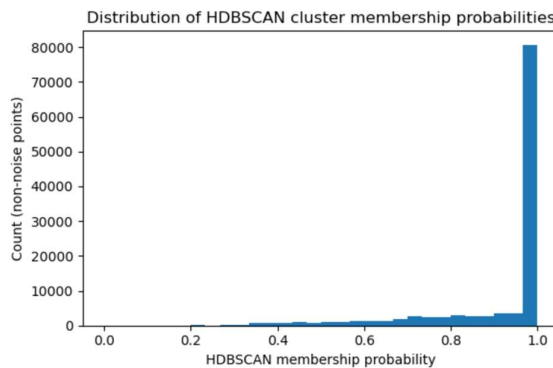


Figure 12. Distribution of HDBSCAN membership probabilities (non-noise points). Most points are assigned with high confidence, with a large concentration at 1.0. This indicates strong cluster identification across the majority of clusters.

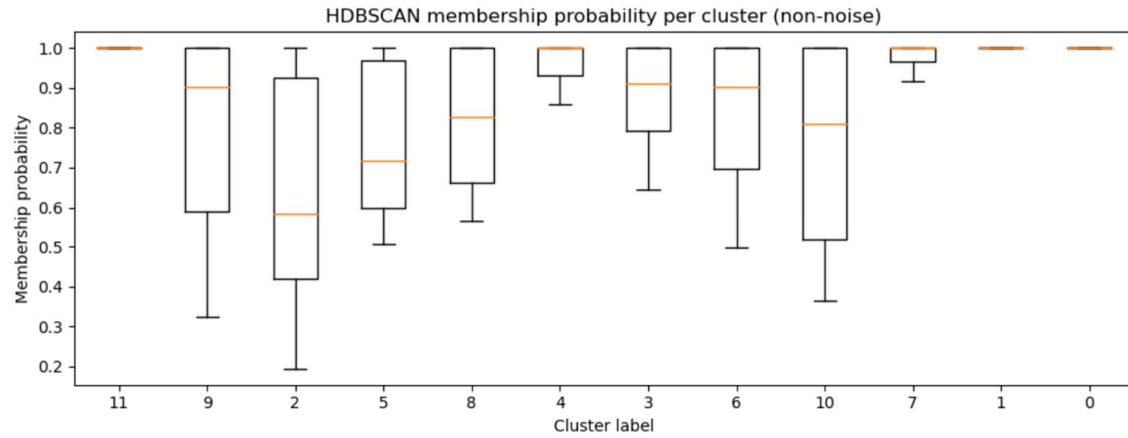


Figure 13. HDBSCAN membership probability by cluster number. Steady-state clusters (11, 8, 3, 7, 1) and depressurised (0) regimes show high cluster confidence. Anomalies or loss-related clusters (2, 5, 6, 10) display wider probability distributions; this may indicate more transitional behaviour with less certainty.